3 **(a)** **(i)** resonance B1 [1]

(ii) amplitude 16 mm <u>and</u> frequency 4.6 Hz A1 [1]

(b) **(i)** $a = (-)\omega^2 x$ and $\omega = 2\pi f$ C1
$a = 4\pi^2 \times 4.6^2 \times 16 \times 10^{-3}$ C1
$= 13.4\,\text{m s}^{-2}$ A1 [3]

(ii) $F = ma$ C1
$= 150 \times 10^{-3} \times 13.4$
$= 2.0\,\text{N}$ A1 [2]

(c) line always 'below' given line and never zero M1
peak is at 4.6 Hz (or slightly less) and flatter A1 [2]

4 **(a)** charge / potential (difference) *(ratio must be clear)* B1 [1]